MILITARY SPECIFICATION

EYEPIECE, BIOCULAR, AN/VSG-2

This specification is approved for use by USAERADCOM, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers the Eyepiece, Biocular, AN/VSG-2.

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

| | |
|---|---|
| NN-P-71 | - Pallets, Material Handling, Wood, Stringer Construction, 2-way and 4-way (Partial) |
| QQ-S-781 | - Strapping, Steel, and Seals |
| NNN-P-40 | - Paper, Lens |
| PPP-B-566 | - Boxes, Folding, Paperboard |
| PPP-B-585 | - Boxes, Wood, Wirebound |
| PPP-B-601 | - Boxes, Wood, Cleated-Plywood |
| PPP-B-621 | - Boxes, Wood, Nailed and Lock-Corner |
| PPP-B-636 | - Boxes, Shipping, Fiberboard |
| PPP-B-640 | - Boxes, Fiberboard, Corrugated, Triple-Wall |
| PPP-C-795 | - Cushioning Material, Flexible, Cellular, Plastic Film for Packaging Applications |
| PPP-F-320 | - Fiberboard, Corrugated and Solid, Sheet Stock (Container Grade), and Cut Shapes |
| PPP-S-760 | - Strapping, Nonmetallic (and Connectors) |
| PPP-T-45 | - Tape, Gummed, Paper, Reinforced and Plain, for Sealing and Securing |
| PPP-T-97 | - Tape, Packaging/Industrial, Filament Reinforced |

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Hq, USA Electronics Research and Development Command, ATTN: DRDEL-E, 2800 Powder Mill Road, Adelphi, MD 20783 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 5855

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MILITARY

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|---|---|
| MIL-P-116 | - Preservation-Packaging, Methods of |
| MIL-P-11268 | - Parts, Materials, and Processes used in Electronic Equipment |

STANDARDS

MILITARY

| | |
|---|---|
| MIL-STD-105 | - Sampling Procedures and Tables for Inspection by Attributes |
| MIL-STD-129 | - Marking for Shipment and Storage |
| MIL-STD-147 | - Palletized and Containerized Unit Loads, 40 Inch x 48 Inch Pallets, Skids, Runners, or Pallet-Type Base |
| MIL-STD-150 | - Photographic Lenses |
| MIL-STD-252 | - Classification of Visual and Mechanical Defects for Equipment, Electronic, Wired, and Other Devices |
| MIL-STD-454 | - Standard General Requirements for Electronic Equipment |
| MIL-STD-810 | - Environmental Test Methods |
| MIL-STD-1188 | - Commercial Packaging of Supplies and Equipment |

FEDERAL

| | |
|---|---|
| FED-STD-595 | - Colors |

DRAWINGS

USA ELECTRONICS COMMAND

| | |
|---|---|
| DLSM-B-805103 | - Eyepiece, Biocular, AN/VSG-2 |

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

## 3. REQUIREMENTS

3.1 Item definition. The Eyepiece, Biocular, AN/VSG-2, herein known as the biocular eyepiece, magnifies the visual image at the output of the image intensifier and presents the scene to the eye of the viewer at the gunner's or commander's station.

3.2 Construction. The biocular eyepiece shall be constructed in accordance with DLSM-B-805103 and as specified herein.

3.2.1 Weight. The total weight of the biocular eyepiece shall not exceed 10 pounds.



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3.3 First article. When specified in the contract (see 6.2), the contractor shall furnish first article units for inspection.

3.4 Materials, parts, and processes. Materials, parts, and processes shall be as specified herein and as shown on the applicable drawings. Materials, parts, and processes not specified shall be selected by the contractor in accordance with MIL-P-11268.

3.5 Components. The biocular eyepiece shall consist of optical elements and a housing.

3.6 Performance characteristics.

3.6.1 Field of view to the eye. The field of view to the eye shall not be less than 39 degrees.

3.6.2 Magnification. The on-axis magnification shall not be less than 4.5 times.

3.6.3 Exit pupil, eye relief, and field overlap. The biocular eyepiece shall have a 60-mm exit pupil and a 60-mm eye relief for an on-axis image. Eye relief is measured from the vertex of the last glass surface to a plane perpendicular to the optical axis in which the exit pupil is located. When two 5-mm exit pupils, decentered plus and minus 27.5 mm from the optical axis (55-mm center-to-center separation) are located at the 60-mm eye relief distance, the biocular eyepiece shall provide at least 50-percent field overlap with field overlap calculated using the diameter and not the area of the overlapping images. These conditions shall be satisfied at the fixed diopter setting.

3.6.4 Biocular focus. The eyepiece shall be focussed at -2.5, ±0.3 diopters.

3.6.5 Field flatness. The biocular eyepiece shall have a flat tangential field within plus or minus 1/2 diopter across the outer 50 percent of the 40-mm format and within plus or minus 1/4 diopter across the central 50 percent of the 40-mm format at the fixed diopter setting.

3.6.6 Resolution. The visual axial resolution (central 25 percent of the field of view) of the biocular eyepiece shall be not less than 35.9 line pairs per millimeter (Group 5, Element 2 of a 1951 Air Force Resolution Target). The remaining field of view shall resolve 17.9 line pairs per millimeter (Group 4, Element 2).

3.6.7 Transmission. The transmission of the biocular eyepiece shall be at least 85 percent over the entire aperture for the region of spectral output of a P-20 phosphor.

3.6.8 Effective focal length (EFL). The effective focal length shall be 1.733, ±0.017 inches.

3.6.9 Modulation transfer function (MTF). With a single 5-mm exit pupil decentered plus or minus 27.5-mm from the optical axis and located at a 60-mm eye relief, the MTF of the biocular eyepiece shall be at least equal to the



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MTF given below. This MTF shall be measured at the fixed diopter setting for an image placed at the center of the format. Axial MTF values under these conditions shall be not less than the following:

| Frequency (lp/mm) | MTF |
|---|---|
| 2 | 0.97 |
| 4 | 0.94 |
| 6 | 0.91 |
| 8 | 0.85 |
| 10 | 0.78 |

3.6.10 Linear distortion. The linear distortion of the biocular eyepiece measured at the fixed diopter setting shall be 5.3, ±1.0 percent barrel distortion at the edge of the 40-mm format.

3.6.11 Back focal distance. The back focal distance shall not exceed 0.10 inch for an object projected 400 mm behind (towards the biocular) for the nominal 60-mm exit pupil.

3.6.12 Sealing. The biocular eyepiece shall be assembled in a dry nitrogen atmosphere. The eyepiece shall then be purged with nitrogen that is capable of withstanding temperatures down to -57°C without producing condensation. The eyepiece shall then be sealed.

3.6.13 Cleanliness. The biocular eyepiece shall contain no foreign matter such as dust, dirt, fingerprints, moisture, etc., that can be detected by visual examination.

3.7 Environmental conditions.

3.7.1 Temperature shock. The biocular eyepiece shall not be damaged (see 6.3) by sudden changes in temperature between -57°C to +71°C.

3.7.2 High temperature. The biocular eyepiece shall not be damaged by storage to +71°C and operation to +52°C.

3.7.3 Low temperature. The biocular eyepiece shall not be damaged by operation to -32°C and storage to -57°C.

3.7.4 Shock. With the eyepiece mounted in a fixture, it shall be subjected to the following shocks:

a. Basic design: 40 g's, 18 milliseconds, 1/2 sine, 18 shocks total for operation.

b. High intensity: 100 g's, 4 milliseconds, 1/2 sine, 18 shocks total for operation.

c. Bench handling: The biocular eyepiece shall not be damaged by bench handling during normal activities of operation, shipping, and storage.



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3.7.5 Vibration. The biocular eyepiece shall not be damaged when subjected to the following random and superimposed sinusoidal levels for a period of 15 minutes in each of three mutually perpendicular axes. (Group C: 5 minutes per axis.)

Random: $6 \times 10^{-3}$ $g^2$/Hz, 5 to 500 Hz

Sine: 1.06 g's, X axis
1.51 g's, Y axis
1.06 g's, Z axis

Octave 1 42 to 84 Hz

Octave 2 84 to 168 Hz

3.7.6 Altitude. The biocular eyepiece shall operate at altitudes up to 10,000 feet above sea level. The biocular eyepiece shall operate after exposure to altitudes up to 40,000 feet above sea level.

3.8 Treatment and painting. Unless otherwise specified, the eyepiece and its components and parts shall be cleaned, treated, and painted in accordance with the applicable drawing. The color of the biocular eyepiece shall be tank white, No. 17875 of FED-STD-595.

3.9 Nameplates and product marking. Unless otherwise specified, the eyepiece, parts, components, subassemblies, and assemblies thereof shall be marked for identification and reference designation markings in accordance with the applicable drawing.

3.10 Workmanship. Workmanship shall be in accordance with MIL-STD-454, Requirement 9.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspections. Inspections shall be classified as follows:

a. First article inspection (see 4.3).
b. Quality conformance inspection (see 4.5).
c. Inspection of packaging (see 4.8).



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4.3 First article inspection. Unless otherwise specified in the contract or purchase order, first article inspection shall be performed by the contractor.

4.3.1 Inspections. All materials, parts, processes, and assemblies shall be examined for conformance to the applicable specification or drawing. Inspections shall be made using MIL-STD-105, General Inspection Level II with a 2.5 percent AQL for mechanical and a 4 percent AQL for visual inspection, except as noted in table I.

TABLE I. Inspections.

| Inspection Requirement | Percent AQL | Requirement paragraph | Inspection criteria |
|---|---|---|---|
| Any part or component missing or damaged | 2.5 | 3.2, 3.4 | MIL-STD-252 |
| Weight not as specified | 2.5 | 3.2.1 | MIL-STD-252 |
| Marking not as specified | 2.5 | 3.2 | |
| Workmanship not as specified | 2.5 | 3.10 | MIL-STD-454 |
| Treatment and painting not as specified | 2.5 1/ | 3.2 | |
| Dimensions not as specified | 2.5 | 3.2 | |
| Cleanliness | 100 | 3.6.13 | MIL-STD-252 |

1/ MIL-STD-105, Inspection Level S-3 or objective quality evidence.

4.3.2 Tests. Upon successful completion of inspections specified in 4.3.1, first article biocular eyepieces shall be subjected to the tests in table II. Tests shall be conducted in the order listed. Failure of any tests shall constitute first article failure.

TABLE II. First article inspection.

| Inspection | Requirement paragraph | Test paragraph |
|---|---|---|
| Field of view to the eye | 3.6.1 | 4.6.1 |
| Magnification | 3.6.2 | 4.6.2 |
| Exit pupil, eye relief, and field overlap | 3.6.3 | 4.6.3 |
| Biocular focus | 3.6.4 | 4.6.4 |
| Field flatness | 3.6.5 | 4.6.5 |
| Resolution | 3.6.6 | 4.6.6 |
| Transmission | 3.6.7 | 4.6.7 |
| Effective focal length | 3.6.8 | 4.6.8 |
| Modulation transfer function | 3.6.9 | 4.6.9 |
| Linear distortion | 3.6.10 | 4.6.10 |
| Back focal distance | 3.6.11 | 4.6.11 |
| Temperature shock | 3.7.1 | 4.7.1 |



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TABLE II. First article inspection. - Continued

| Inspection | Requirement paragraph | Test paragraph |
|---|---|---|
| High temperature | 3.7.2 | 4.7.2 |
| Low temperature | 3.7.3 | 4.7.3 |
| Shock | 3.7.4 | 4.7.4 |
| Vibration | 3.7.5 | 4.7.5 |
| Altitude | 3.7.6 | 4.7.6 |

4.4 Inspection procedures. Unless otherwise specified herein, the biocular eyepiece shall be tested at an ambient temperature of 23°, ±5°C.

4.5 Quality conformance inspection.

4.5.1 Inspection. Inspection shall be as specified in 4.3.1.

4.5.2 Tests.

4.5.2.1 Group A inspection. Group A inspection shall be conducted on eyepieces selected from units which have passed the inspection specified in 4.5.1. Group A tests listed in table III may be performed in any order. Failure of any test shall be cause for rejection of that unit.

TABLE III. Group A inspection.

| Inspection | Requirement paragraph | Test paragraph |
|---|---|---|
| Exit pupil, eye relief, and field overlap | 3.6.3 | 4.6.3 |
| Biocular focus | 3.6.4 | 4.6.4 |
| Resolution | 3.6.6 | 4.6.6 |

4.5.2.2 Group B inspection. Group B inspection shall be conducted on biocular eyepieces selected from units which have passed the tests in 4.5.2.1. Sampling of lots (see 6.2) shall be in accordance with MIL-STD-105, AQL 2.5 percent defective, inspection level S3 combined. Group B tests listed in table IV may be performed in any order.

TABLE IV. Group B inspection.

| Inspection | Requirement paragraph | Test paragraph |
|---|---|---|
| Transmission | 3.6.7 | 4.6.7 |
| MTF | 3.6.9 | 4.6.9 |
| Back focal distance | 3.6.11 | 4.6.11 |



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4.5.2.3 Group C inspection. Group C inspections shall be conducted on eyepieces selected from units which have passed the tests in 4.5.2.1. The sample(s) shall be tested in accordance with the inspections listed in table V. Samples shall be selected in accordance with 4.5.2.3.1. Group C tests listed in table V shall be performed in the order listed.

4.5.2.3.1 Sampling for group C inspection.

4.5.2.3.1.1 Subgroup 1. For subgroup 1, three biocular eyepieces shall be selected from the first quality conformance inspection lot. For subsequent group C inspection, three biocular eyepieces shall be selected every month from the monthly specified quantity (see 6.2).

4.5.2.3.1.2 Subgroup 2. For subgroup 2, one biocular eyepiece shall be selected from every sixth quality conformance inspection lot, starting with the first lot.

TABLE V. Group C inspection.

| Inspection | Requirement paragraph | Test paragraph |
|---|---|---|
| Subgroup 1 | | |
| Field of view to the eye | 3.6.1 | 4.6.1 |
| Magnification | 3.6.2 | 4.6.2 |
| Field flatness | 3.6.5 | 4.6.5 |
| Effective focal length | 3.6.8 | 4.6.8 |
| Linear distortion | 3.6.10 | 4.6.10 |
| Subgroup 2 | | |
| Temperature shock | 3.7.1 | 4.7.1 |
| High temperature | 3.7.2 | 4.7.2 |
| Low temperature | 3.7.3 | 4.7.3 |
| Shock | 3.7.4 | 4.7.4 |
| Vibration | 3.7.5 | 4.7.5 |
| Altitude | 3.7.6 | 4.7.6 |

4.5.2.3.2 Group C failures. Actions required relative to group C failure shall be as specified in the contract (see 6.2).

4.6 Test methods. The biocular eyepiece shall be tested in accordance with government approved test procedure. Unless otherwise specified, test shall be conducted at ambient temperature with light source simulating the P-20 phosphor and an S-11 phototube.

4.6.1 Field of view to the eye. This test is part of the linear distortion test given in 4.6.10. Failure to meet the requirements of 3.6.1 shall constitute failure of the test.



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4.6.2 Magnification. This test is part of the linear distortion test given in 4.6.10. Failure to meet the requirement of 3.6.2 shall constitute failure of this test.

4.6.3 Exit pupil, eye relief, and field overlap. The test apparatus shall be arranged as shown in figure 1. The special test pattern is placed in the focal plane of the biocular eyepiece so that a diopter scope centered at a 60-mm eye relief sees the virtual image clearly when set at minus 2.5, ±0.3 diopters. Remove the diopter scope and place a physical stop at the 60-mm eye relief distance that has two 5.0, ±0.1-mm-diameter pupils marked F and G decentered plus and minus 27.5 mm from the optical axis with the line joining the two pupils being parallel to the line joining the pinholes in the pattern. Look through pinhole F in the exit pupil stop and determine if pinholes B, C, D, and E can be observed. Look through pinhole G and determine if pinholes A, B, C, and D can be observed. The specified pinholes shall be observed. Failure to meet the requirements of 3.6.3 shall constitute failure of this test.

4.6.4 Biocular focus. This is a part of the linear distortion test given in 4.6.10. Failure to meet the requirement of 3.6.4 shall constitute failure of this test.

4.6.5 Field flatness. This test is part of the linear distortion test given in 4.6.10. Failure to meet the requirement of 3.6.5 shall constitute failure of this test.

4.6.6 Resolution. Place a positive high contrast USAF 1951 resolution test target in the proper focal plane of the biocular eyepiece with the 5 group centered in the format. Back illuminate the test target with Lambertian P-20 (green) light of sufficient intensity so that the clear portions of the test target are at 30 foot-lamberts brightness. The proper focal plane is the focal plane for which the virtual image of the test target is set at minus 2.5, ±0.3 diopters when viewed through the biocular eyepiece with a diopter scope placed at a 60 mm eye relief. View the test target biocularly without a visual aid and read the maximum resolvable pattern. This pattern must be Group 5, Element 2. Repeat this procedure for the edge of the field where the observer must resolve Group 4, Element 2. Failure to meet the requirements of 3.6.6 shall constitute failure of this test.

4.6.7 Transmission. The transmission of the biocular eyepiece using P-20 light and an S-11 photodetector shall be measured according to Method 6 of MIL-STD-150 with the following modification. The aperture of the collimator shall be 25 mm and the transmission is the ratio of the photodetector reading when light passes through the biocular eyepiece into the integrating sphere compared to the photodetector reading when the light does not pass through the biocular eyepiece into the sphere. Failure to meet the requirements of 3.6.7 shall constitute failure of this test.

4.6.8 Effective focal length. The effective focal length shall be measured in green light as specified in MIL-STD-150. Failure to meet the requirements of 3.6.8 shall constitute failure of this test.



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4.6.9 Modulation transfer function. The MTF of the biocular eyepiece shall be measured on a sine-wave tester that will measure the MTF between 0 and 10 lp/mm in the focal plane of the eyepiece. This measurement may be either continuous or in a minimum of five equally spaced points. The eyepiece shall be measured at the prescribed fixed diopter setting for an on-axis image and a 5-mm exit pupil decentered 27.5 mm from the optical axis at a 60-mm eye relief. The following test procedure shall be used. Any alternate test procedure must have prior approval of the Government.

1. The basic test setup is diagrammed in figure 2.
2. Measure the length 50 m of the slit and record for future use. (May be done with auxiliary microscope.)
3. Without the biocular eyepiece in place, position the slit 400 mm away from the 50-mm relay lens fitted with a 5-mm stop. Measure the MTF of there lay lens in P-20, light S11 response and calculate the diffraction limit and geometric MTF of this relay (see figure 3).
4. Insert the biocular eyepiece in its proper position (60-mm eye relief and 27.5-mm decentered pupil) and focus the system by moving the slit image for maximum MTF. Record this MTF curve.
5. Next, measure the length of the slit image using the Odetta head or equivalent knife edge scanner and calculate the magnification. Magnification = $\frac{\text{Length of slit image}}{\text{Length of slit object}}$. The frequency scale at the Odetta head or equivalent knife edge scanner is then adjusted to represent the spatial frequency scale in the biocular focal plane.
6. Calculate the biocular eyepiece MTF by using the MTF of the adjusted curve divided by the geometric MTF of the 50-mm relay lens.

Failure to meet the requirements of 3.6.9 shall constitute failure of this test.

4.6.10 Linear distortion. Set up test equipment as shown in figure 4. Back illuminate the test target with lambertian P-20 light. Place the test pattern perpendicularly to the optical axis of the biocular eyepiece, and coplanar with the rotary table axis. Place the biocular eyepiece such that its optical axis is coincident with the optical axis of the diopter scope, and the eye relief is 60 mm. Set the test target so that zero on the test target coincides with zero on the rotary table. Rotate the table (diopter scope about its front aperture) and verify that the 20-mm extremes of the test target are visible. Failure to see both 20-mm extremes constitutes failure to meet the requirements of 3.6.1. Next, start at the +20-mm field height record angle read on the rotary table versus field height in 1-mm increments until -20-mm field height is obtained. At each 1-mm increment of field height, determine the amount of refocus of the diopter scope needed to achieve a sharp image. Failure to meet the requirements of 3.6.5 shall be cause for rejection. Consider the plus and minus 1 mm field height readings and obtain the average tan $\theta_1$ ($\overline{\tan \theta_1}$) of the two output angles $\theta_{+1}$ and $\theta_{-1}$. From this magnification M can be calculated using:

$$M = \frac{254 \text{ mm } \overline{\tan \theta_1}}{1 \text{ mm}}$$



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Record the angular readings of the rotary table to verify compliance with 3.6.1. Failure to meet the requirement of 3.6.2 shall be cause for rejection. Percent distortion, $\%D_F$, at any field height is calculated as:

$$\%D_F = \left[ \frac{K \overline{\tan \theta}_F}{MF} - 1 \right] 100$$

where

$K = 254$ mm

$F$ = field height in mm

$\overline{\tan \theta}_F$ = average tan of $\theta_{+F}$ and $\theta_{-F}$

$M$ = on-axis magnification.

Passage of all portions of this test procedure will assure that the requirements of 3.6.4 have been met. Failure to meet the requirement of 3.6.10 shall constitute failure of this test.

4.6.11 Back focal distance. Using a well corrected lens, form an image 400 mm behind the nominal biocular entrance pupil. This image will act as the biocular's object. The back focal distance (without a high voltage shield) shall be less than 0.1 inch.

4.6.12 Sealing. The biocular eyepiece shall be examined for conformance to the requirement.

4.7 Environmental tests. The following environmental tests shall be conducted as specified in table II with the biocular eyepiece mounted on the TTS or on a test fixture which simulates its normal operational mounting configuration, and protects those portions of the biocular eyepiece which are not normally exposed to the environment. Performance tests 4.6.1 through 4.6.11 shall be run after completion of all environmental tests.

4.7.1 Temperature shock. The test shall be conducted in accordance with MIL-STD-810, Method 503, Procedure I. Failure to meet the requirements of 3.7.1 shall constitute failure of this test.

4.7.2 High temperature. Subject the assembly to the high temperature test specified in MIL-STD-810, Method 501, Procedure II, except that storage temperature of step 4 shall be 71°C. At the conclusion of step 4, stabilize the assembly at 52°C and subject it to the performance tests of 4.6.6. Failure to meet the requirements of the performance tests and the requirements of 3.7.2 shall constitute failure of this test.

4.7.3 Low temperature. Subject the biocular eyepiece to the low temperature test specified in MIL-STD-810, Method 502, Procedure I. Stabilize the temperature at -57°C for a period of 12 hours minimum. At the conclusion of this time, stabilize the assembly at -32°C and subject it to the performance tests of 4.6.6. Failure to meet the requirements of the performance tests and the requirements of 3.7.3 shall constitute failure of this test.



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4.7.4 Shock.

4.7.4.1 Basic design. Subject the assembly to the test specified in MIL-STD-810, Method 516.1, Procedure I for three half-sine wave shock pulses applied in both directions along three mutually perpendicular axes. The environment and application of the shock pulses are shown in 3.7.4. During shock pulses, verify that intermittence does not exist.

4.7.4.2 High intensity. Subject the assembly to the test specified in MIL-STD-810, Method 516.1, Procedure IV for three half-sine wave shock pulses applied in both directions along three mutually perpendicular axes. The environment and application of the shock pulses are shown in 3.7.4. During shock pulses, verify that intermittence does not exist.

4.7.4.3 Bench handling. Subject the biocular eyepiece to the test specified in MIL-STD-810, Method 516.1, Procedure V. At the conclusion of these tests, subject the assembly to the performance tests of 4.6.6.

Failure to meet requirements of 3.7.4 shall constitute failure of these tests.

4.7.5 Vibration. The biocular eyepiece shall be subjected to the vibration test specified in MIL-STD-810, Method 514.1, Procedure VIII, and procedures as modified herein. Mount the biocular eyepiece on the vibration table with the biocular eyepiece non-operating, vibrate in each of three mutually perpendicular axes to the random and superimposed sinusoidal levels of 3.8.6 for 15 minutes. (Group C: 5 minutes per axis.) The appropriate periodic component will be swept up each octave in one minute and down the octave the next minute. Failure to meet the requirements of 3.8.6 shall constitute failure of this test.

4.7.6 Altitude. Place the assembly in an altitude chamber and subject it to the altitude test specified in MIL-STD-810, Method 500, Procedure I, except that:

a. The assembly, in a nonoperating condition, shall be exposed to an atmospheric pressure equivalent to 40,000 feet above sea level for a period of not less than 1 hour.

b. With the chamber pressure increased to 10,000 feet above sea level, the assembly shall be subjected to the performance tests of 4.6.6.

Failure to meet requirements of 3.7.6 shall constitute failure of these tests.

4.8 Inspection of packaging. Packaging shall be inspected in accordance with MIL-P-116 to determine compliance with the requirements of section 5.

5. PACKAGING

5.1 Preservation. Preservation shall be level A, B, or commercial, as specified (see 6.2).

5.1.1 Level A.



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5.1.1.1 Cleaning. Eyepiece, Biocular AN/VSG-2 shall be cleaned in accordance with process C-1 of MIL-P-116.

5.1.1.2 Drying. Eyepiece, Biocular AN/VSG-2 shall be dried in accordance with the applicable procedure of MIL-P-116.

5.1.1.3 Preservative application. None required.

5.1.1.4 Unit packing. Unit packing shall be in accordance with the methods prescribed in MIL-P-116 as specified herein.

5.1.1.4.1 Technical literature. Each technical literature shall be unit packed as specified in MIL-P-116, method IC-1.

5.1.1.4.2 Eyepiece, Biocular AN/VSG-2. Each unit shall be individually unit packed, method III as follows: wrap each assembly with paper, lens conforming to NNN-P-40, type II and secure with tape conforming to PPP-T-45, type III. Cushion each wrapped item with material conforming to PPP-C-795, Class 2, medium thickness. Place each cushioned item within a close-fitting fiberboard box conforming to PPP-B-566, variety 2. Closure shall be in accordance with the appendix of the box specification.

5.1.1.4.3 Intermediate container. Place 10 unit containers within a close-fitting fiberboard box conforming to PPP-B-636, W5c. Place the technical literature, unit packed as specified in 5.1.1.4.1, on top of the contents directly under the lid of the box. Close the box in accordance with the appendix of the box specification. Intermediate containers shall not exceed a maximum of 40 pounds net weight, and a maximum of 1.5 cubic feet with at least two dimensions each not exceeding 16 inches.

5.1.2 Level B. Cleaning, drying, preservative application and unit packing shall be as specified in 5.1.1.

5.1.3 Commercial preservation. Preservation shall be in accordance with MIL-STD-1188.

5.2 Packing. Packing shall be level A, B or commercial, as specified (see 6.2). Shipping containers for level A and B shall be capable of stacking and supporting superimposed loads during shipment and storage without damaging the container(s) or its contents.

5.2.1 Level A. A quantity of Eyepiece, Biocular AN/VSG-2, unit packed as specified in 5.1, shall be packed within a close-fitting box conforming to PPP-B-601, overseas type; PPP-B-621, style 4, class 2; or PPP-B-585, style 2 or 3, class 3. When the gross weight exceeds 200 pounds, or the container length and width is 48 x 24 inches or more and the weight exceeds 100 pounds, 3 x 4 inch skids, laid flat, shall be applied in accordance with the requirements of the container specification, or if not specified in the specification, in a manner which will adequately support the item and facilitate the use of material handling equipment. Closure and strapping shall be in accordance with the applicable container specification or appendix thereto except that metal strapping shall conform to QQ-S-781, type I, finish A.



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5.2.1.1 Unitization. Palletization shall be required when: containers specified in 5.2.1 do not require skids; quantities per destination comprise a pallet load; and container size permits use of one of the pallet patterns of MIL-STD-147. A quantity of containers, packed as specified in 5.2.1, except that container strapping may be omitted, shall be placed on a pallet, load type I, conforming to MIL-STD-147. The pallet shall conform to NN-P-71, type IV, group I or II woods. The load shall be "bonded" to the pallet by strapping conforming to QQ-S-781, type I, finish A, or shrink film.

5.2.2 Level B. A quantity of Eyepiece, Biocular AN/VSG-2, unit packed as specified in 5.1, shall be packed within a close-fitting fiberboard box conforming to PPP-B-640, class 2, style E, or PPP-B-636, type CF, class weather-resistant. The gross weight of boxes conforming to PPP-B-640 shall not exceed 250 pounds. Closure shall be in accordance with the appendix of the applicable box specification. Reinforcing shall be by pressure-sensitive filament tape banding or nonmetallic strapping conforming to PPP-T-97 and PPP-S-760, respectively; selection of the material and application shall be in accordance with the appendix of the applicable box specification. When the gross weight exceeds 200 pounds, or the container length and width is 48 x 24 inches or more and the weight exceeds 100 pounds, containers will be pallet-mounted on pallets conforming to NN-P-71, group I or II woods. The load shall be "bonded" to the pallet by strapping conforming to QQ-S-781, type I, finish A, or shrink film. When a single unit is shipped to a single destination no packing is required; the unit container shall serve as the shipping container. Reinforcing shall be as specified above.

5.2.2.1 Unitization. Palletization shall be required when quantities per destination comprise a pallet load. A quantity of Eyepiece, Biocular AN/VSG-2, unit packed as specified in 5.1, shall be placed on a pallet, load type I, conforming to MIL-STD-147. The pallet shall conform to NN-P-71, type IV, group I or II woods. A fiberboard cap shall be employed over the load having two sides extending down the stacked load at least 12 inches to accommodate marking requirements. The cap shall be fabricated of fiberboard conforming to PPP-F-320, class weather-resistant, W5s or V3c. The load shall be "bonded" to the pallet by strapping conforming to QQ-S-781, type I, finish A, or shrink film.

5.2.3 Commercial packing. Packing shall be in accordance with MIL-STD-1188.

5.3 Marking.

5.3.1 Military marking. In addition to any special marking required by the contract or order, interior packs and exterior shipping containers shall be marked in accordance with MIL-STD-129.

5.3.2 Commercial marking. Marking shall be in accordance with MIL-STD-1188.

6. NOTES

6.1 Intended use. The biocular eyepiece is intended for use with the Tank Mounted Thermal Sight.



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6.2 Ordering data. Procurement documents will specify, as a minimum, the following:

a. Title, number, and date of the development specification.
b. Title, number, and date of this specification.
c. Lot size, if different than specified herein.
d. Requirement to perform first article tests and the quantity of first article test units required in each subgroup.
e. Failure analysis and corrective action requirements for periodic inspection.
f. AQL, if different than specified herein.
g. Requirement for first article and periodic inspection test plans.
h. Disposition of test samples. The approved first article/periodic sample(s) shall be returned to satisfactory operation and submitted for inspection and acceptance as a production item in accordance with section 4.
i. Level A or level B preservation and packing (see section 5).
j. When rough handling test is required.
k. Environmental pollution prevention measures are contained in the packaging material specifications referenced herein. Refer to material specifications or preparing activity for recommended disposability methods.
l. Requirement that commercial packaging shall meet the requirements of MIL-STD-1188, "Alternate requirements" criteria.
m. If biocular eyepieces are procured as part of end item assemblies, group B and group C inspections may be waived except for 4.6.11.
n. MIL-STD-810B shall be used for environmental tests of 4.7.

6.3 Damage. Breakage, loosening, shifting, evidence of corrosion, or failure of any finish, hardware, connection or component, and any degradation of biocular eyepiece performance to values less than specified herein.

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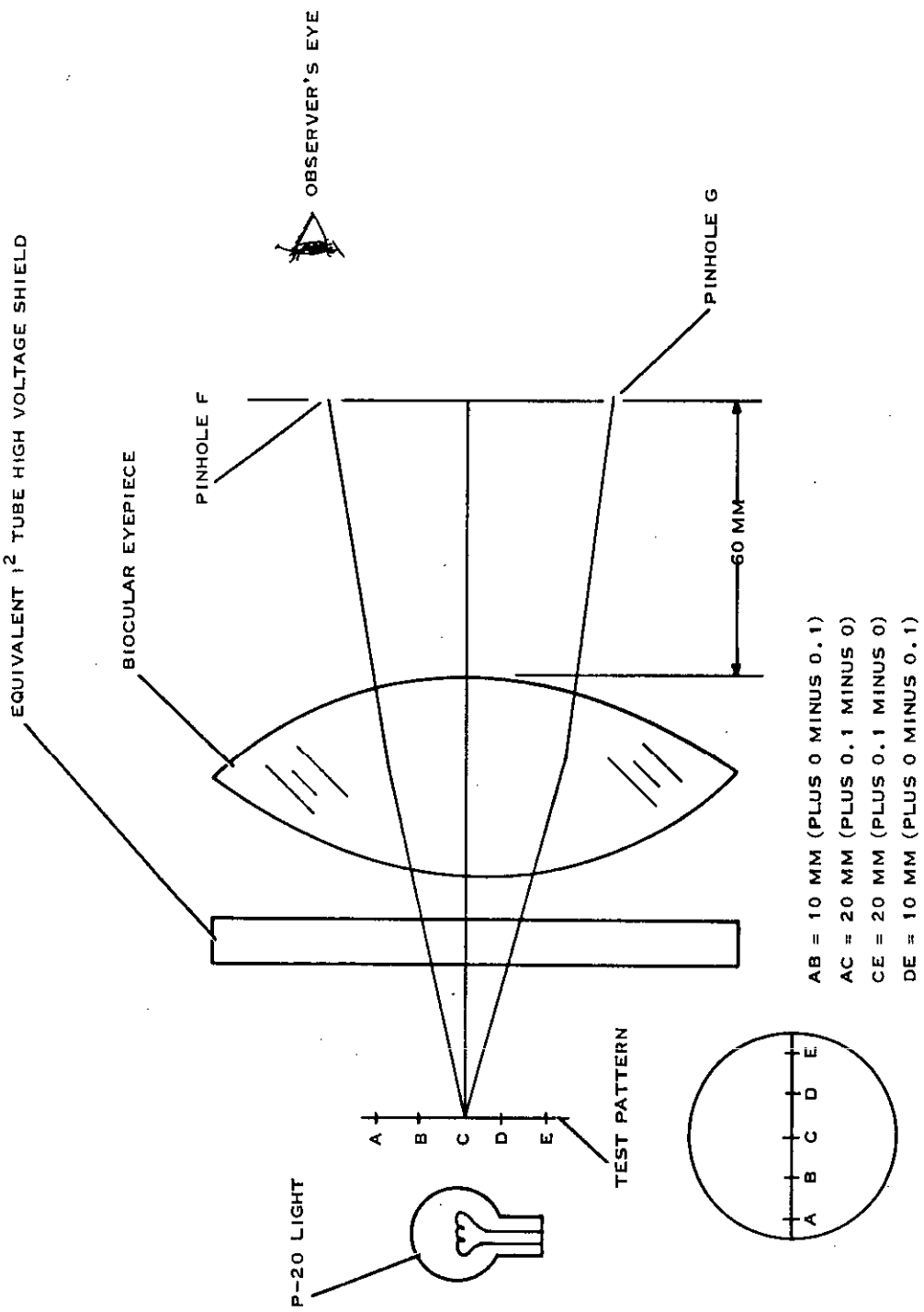



Figure 1. Exit pupil, eye relief, field overlay test configuration.



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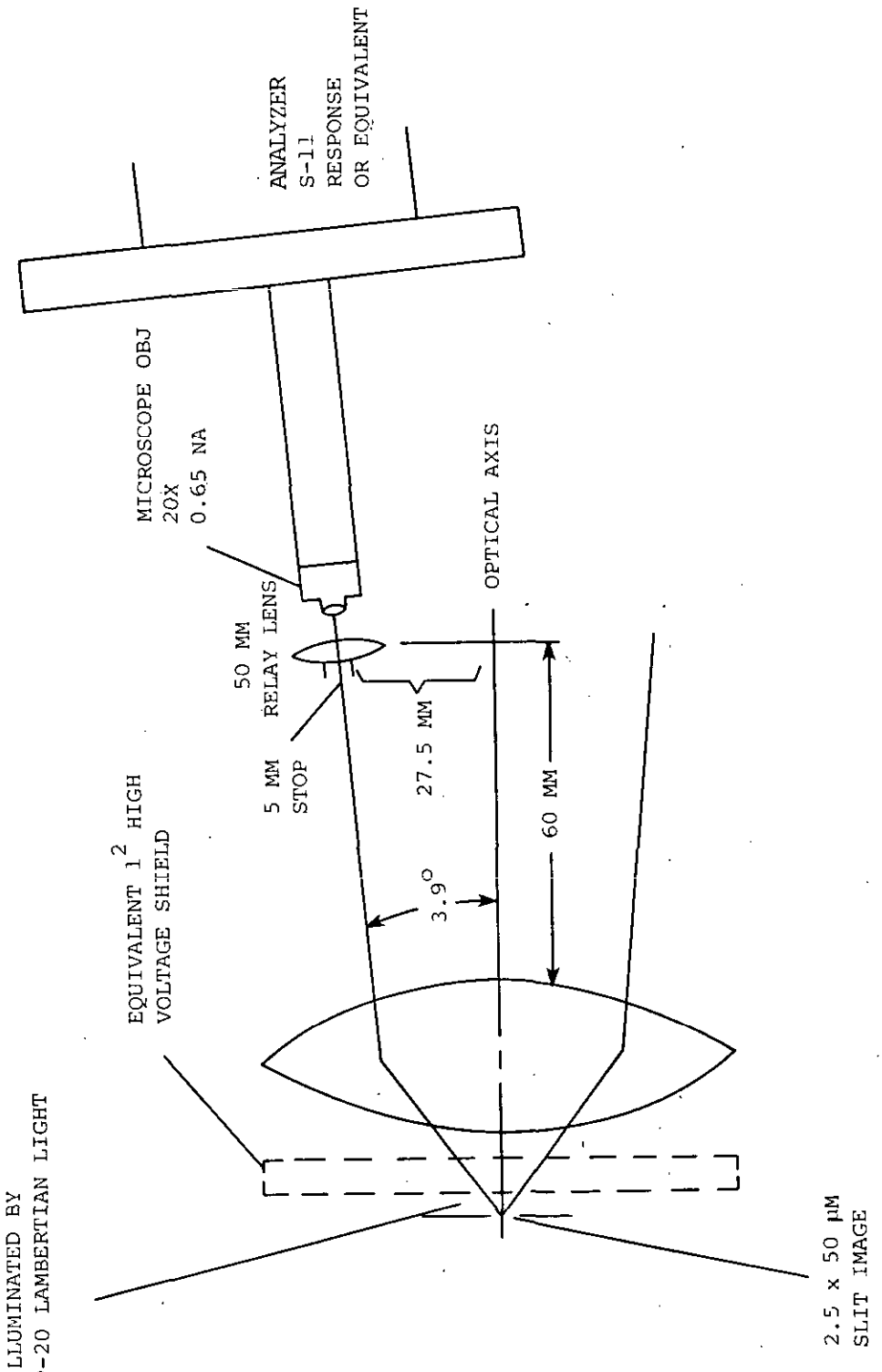



Figure 2. MTF test configuration no. 1.



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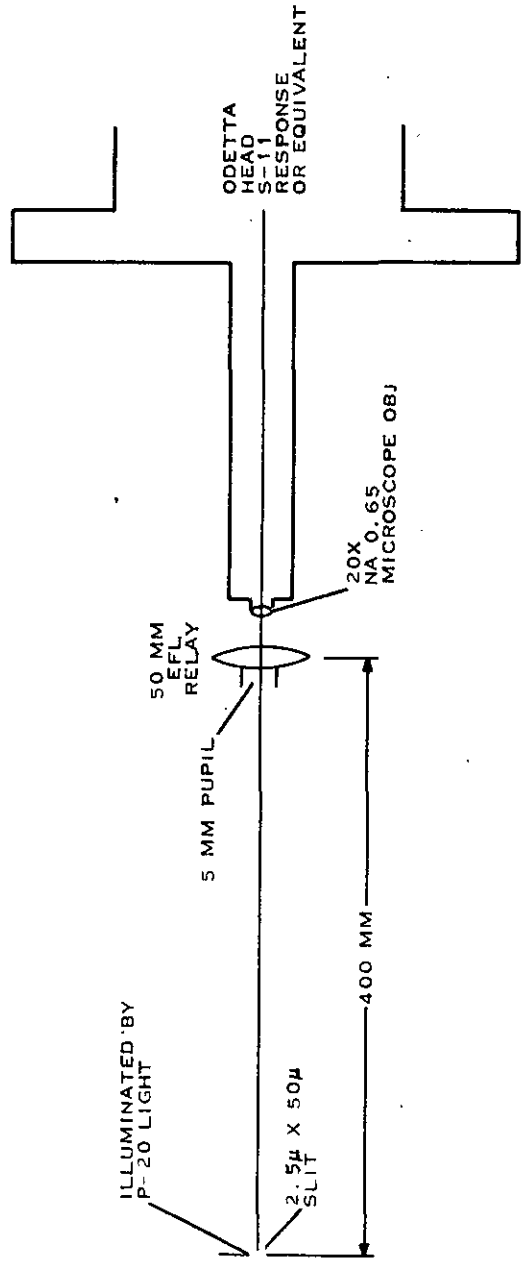



Figure 3. MTF test configuration no. 2.



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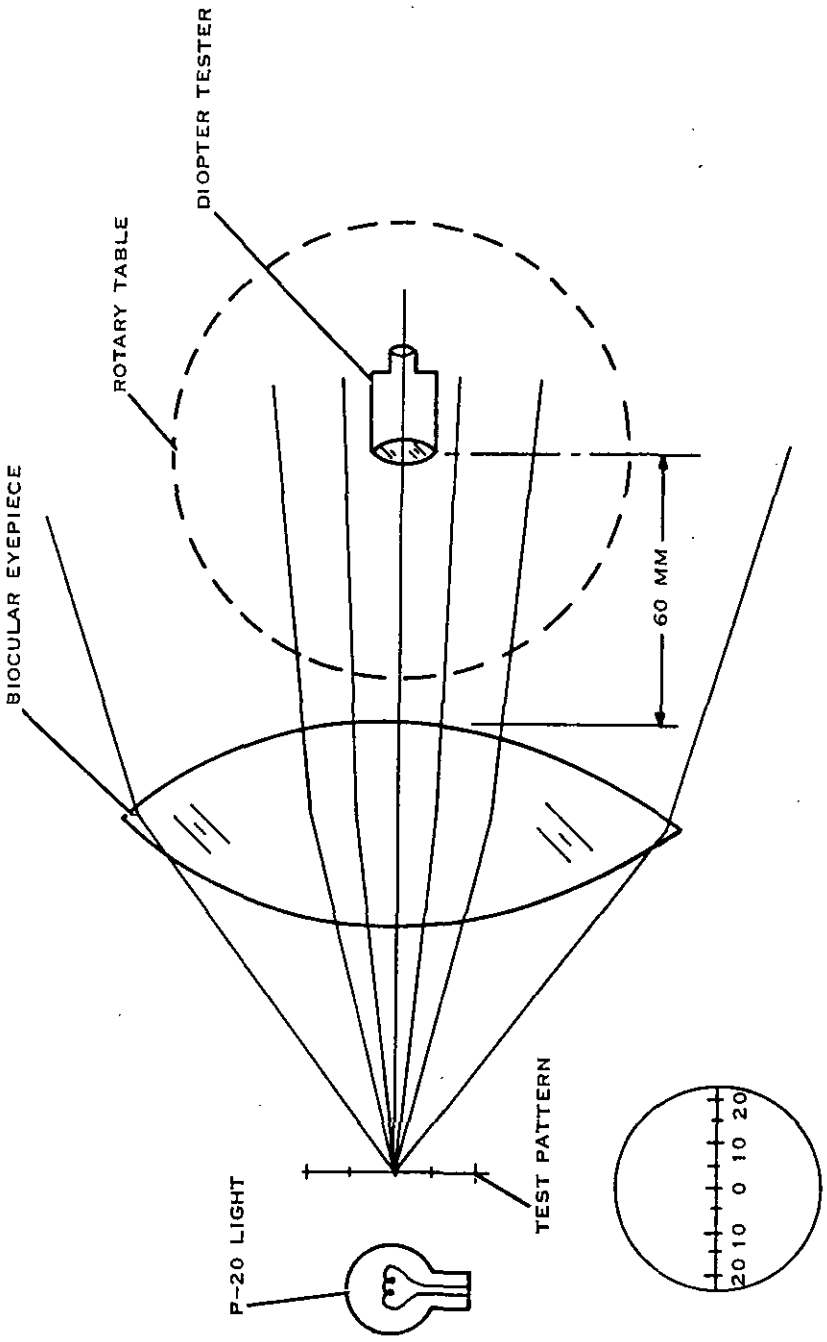



Figure 4. Linear distortion test configuration.



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DOCUMENT IDENTIFIER AND TITLE
MIL-E-49383(ER)

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

   ☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

TELEPHONE NO.

DATE

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